

# Confidence intervals

Estimating population characteristics and expressing uncertainty

Mucyo Karemera · Stéphane Guerrier

Samples and populations

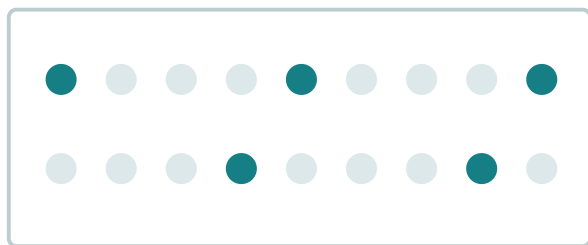
Sampling variability

Confidence intervals

# From our sample to the population

We have described the data in our sample.

Sample



Describe what we observed

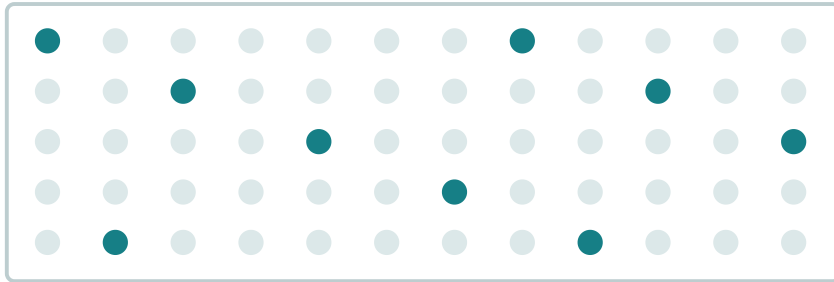
Means, proportions, graphs

These summaries tell us what happened in the sample.

# From our sample to the population

We have described the data in our sample.

Population

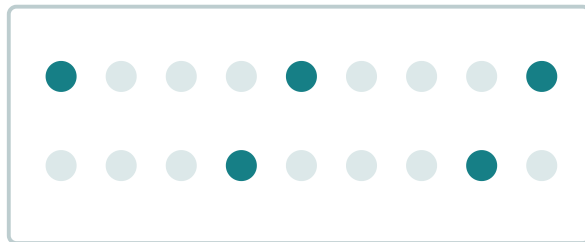


The population we want to learn about  
Beyond the people in our sample



Sampling

Sample



Describe what we observed

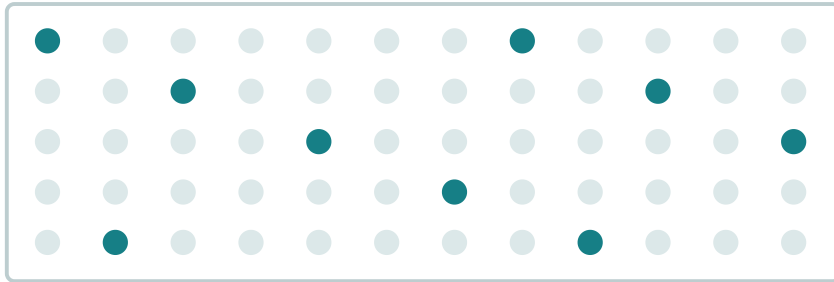
Means, proportions, graphs

Our sample comes from a wider population.

# From our sample to the population

We have described the data in our sample.

Population

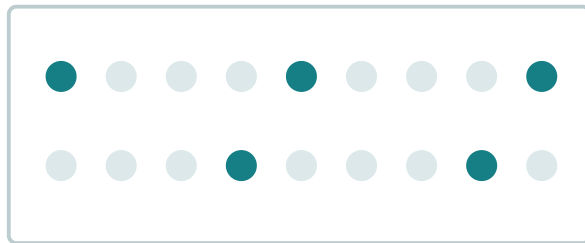


The population we want to learn about  
Beyond the people in our sample



Sampling

Sample



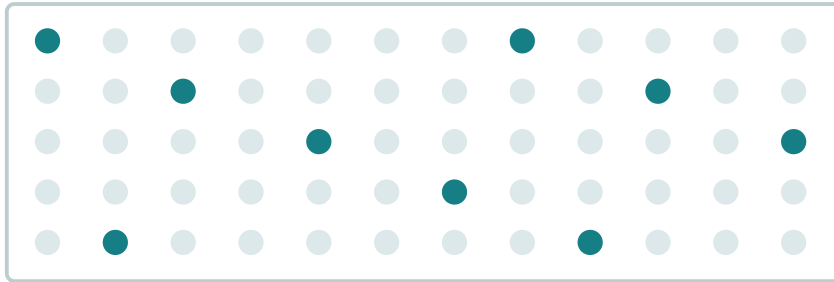
Describe what we observed  
Means, proportions, graphs

What can our sample tell us about that population?

# Descriptive and inferential statistics

Statistical inference connects the sample to the population.

Population

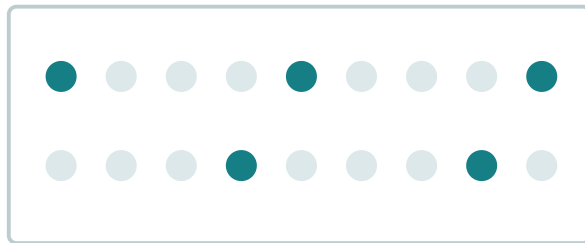


The population we want to learn about  
Beyond the people in our sample



Sampling

Sample



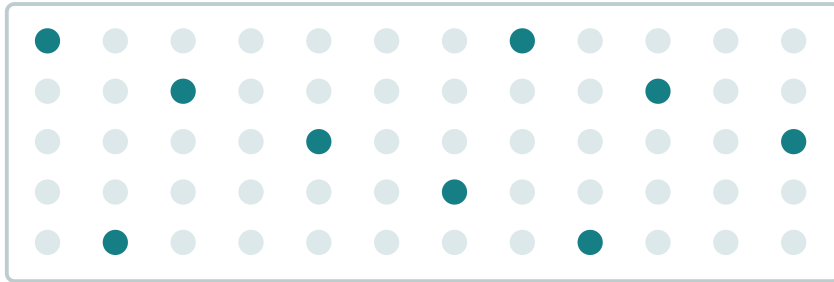
Describe what we observed  
Means, proportions, graphs

A treatment seems to help the patients we studied.

# Descriptive and inferential statistics

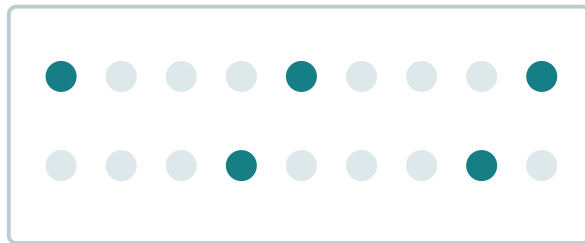
Statistical inference connects the sample to the population.

Population



Sampling

Sample



The population we want to learn about  
Beyond the people in our sample

Infer



Describe what we observed  
Means, proportions, graphs

We estimate population quantities and express our uncertainty.

# Disease prevalence

What proportion of adults in a defined region have the disease?

For this simulation, we choose

$$p = 20\%$$

Knowing the answer lets us study how estimates vary.

Our sampling model

Independent observations.

The same disease probability  $p$  for every observation.

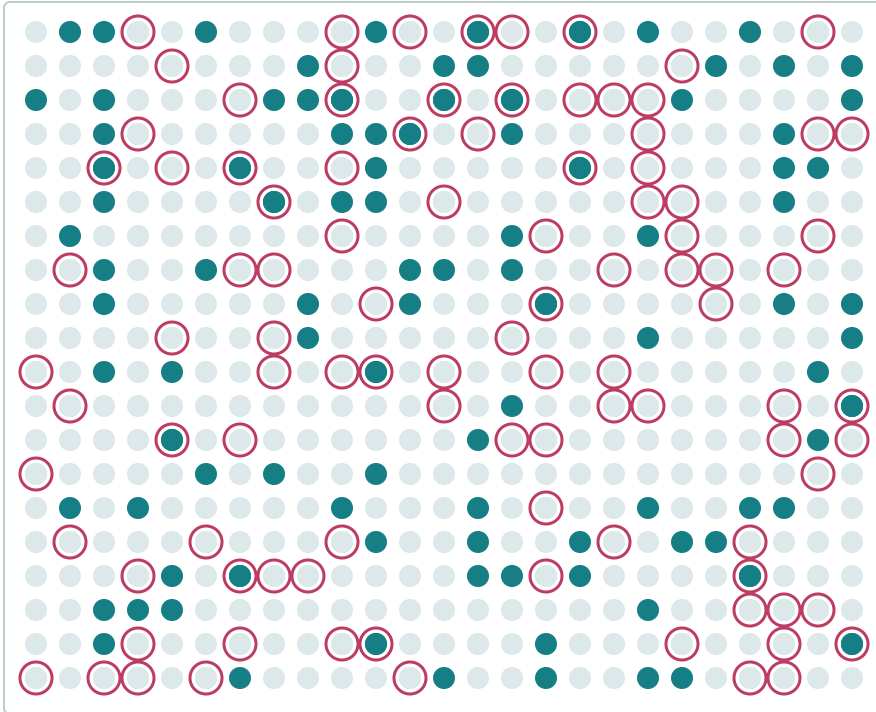
Disease status is recorded accurately.

In a real study, the population prevalence is unknown.

# An observed sample

Population

True prevalence: 20%



Schematic of a large population

Random sample: 100 adults

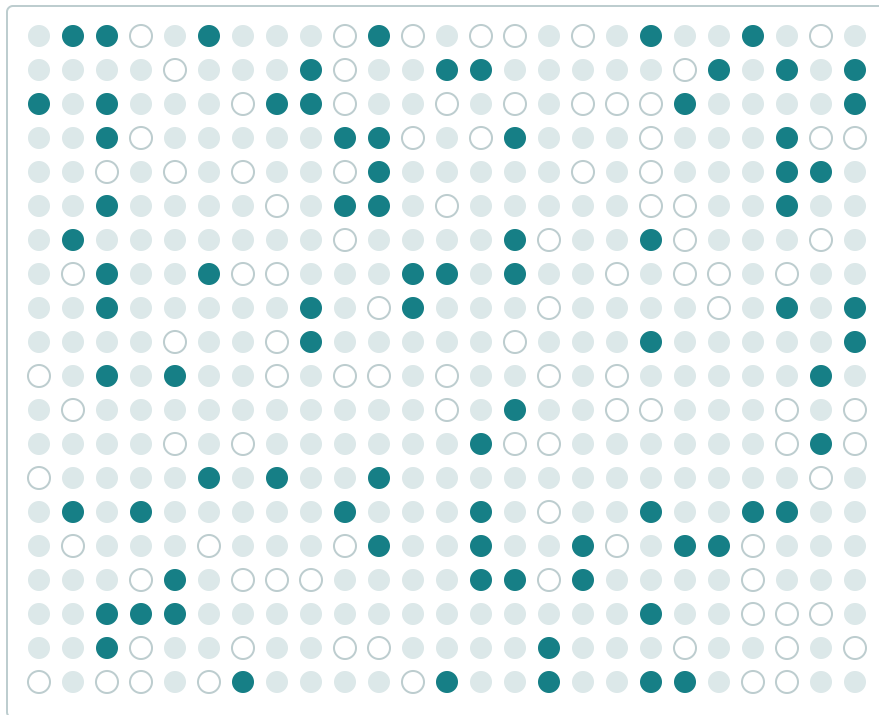
Selection



# An observed sample

Population

True prevalence: 20%

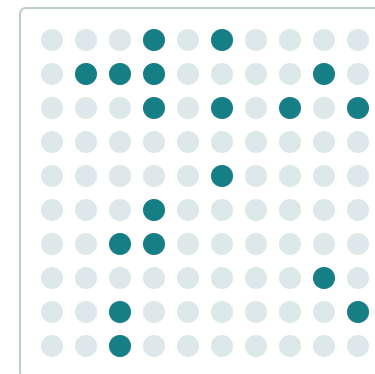


Schematic of a large population

Same individuals



Random sample: 100 adults



18 cases among 100 = 18%

$\hat{p} = 18/100 = 18\%$  An estimate of the population prevalence.

# The estimator and an estimate

An estimator is a rule for calculating a value from sample data.

Disease status:  $1$  = present,  $0$  = absent.

Estimator: the rule for any sample

$$\hat{p} = \frac{X_1 + \cdots + X_n}{n}$$

Number of cases divided by sample size.

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Estimate: the value calculated from our observed sample

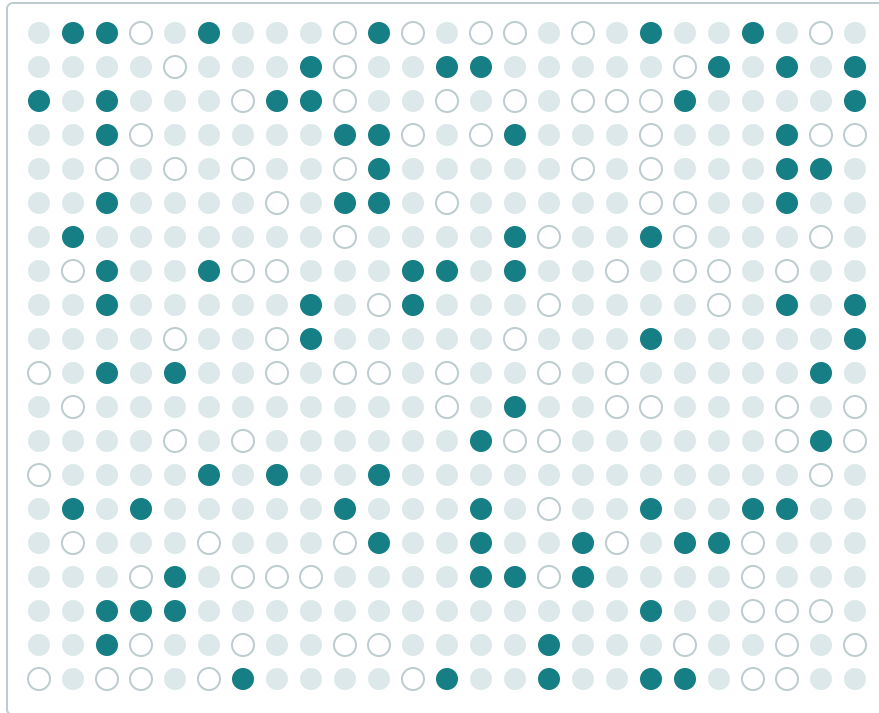
$$\hat{p} = \frac{18}{100} = 18\%$$

For this sample, the estimated prevalence is  $18\%$ .

# Repeated samples from the same population

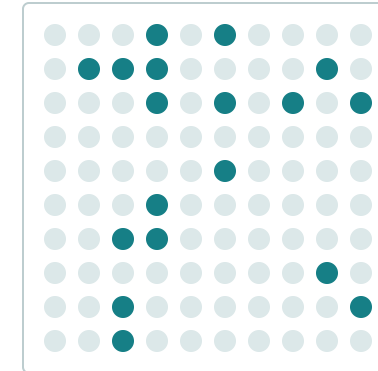
Population

True prevalence: 20%



Schematic of a large population

Random sample: 100 adults



Same individuals



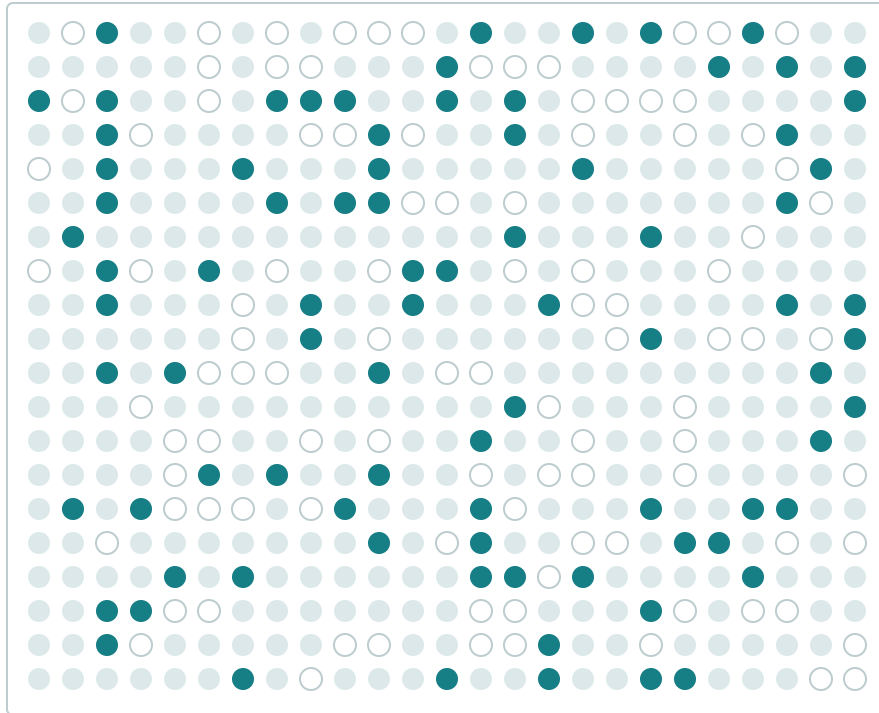
18 cases among 100 = 18%  
Estimates so far: 18%

Same population. Same sample size. The estimate can change.

# Repeated samples from the same population

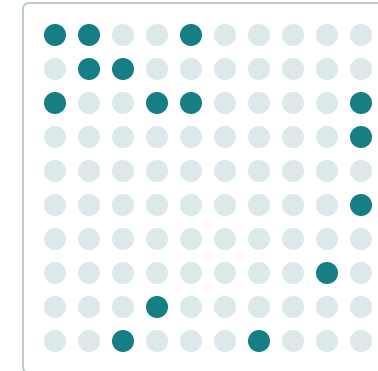
Population

True prevalence: 20%



Schematic of a large population

Random sample: 100 adults



Same individuals



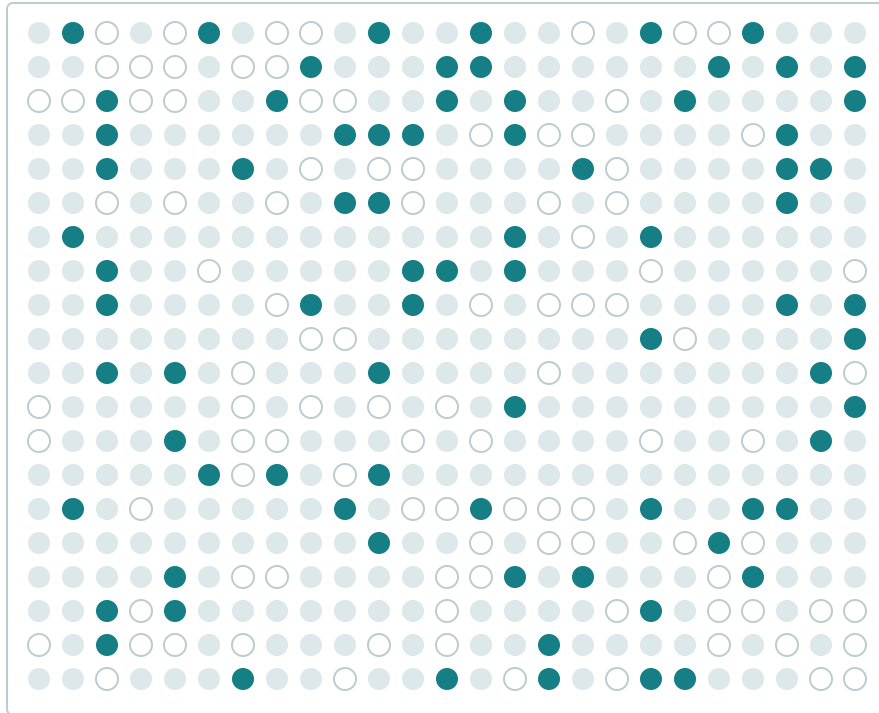
15 cases among 100 = 15%  
Estimates so far: 18% 15%

Same population. Same sample size. The estimate can change.

# Repeated samples from the same population

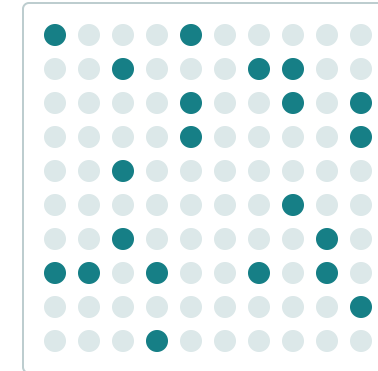
Population

True prevalence: 20%



Schematic of a large population

Random sample: 100 adults



Same individuals



21 cases among 100 = 21%  
Estimates so far: 18% 15% 21%

How far might an estimate fall from the true prevalence of 20%?

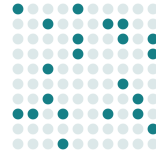
# The distribution of sample proportions

The same process continues: one sample, one estimate.

Population ( $p = 20\%$ )



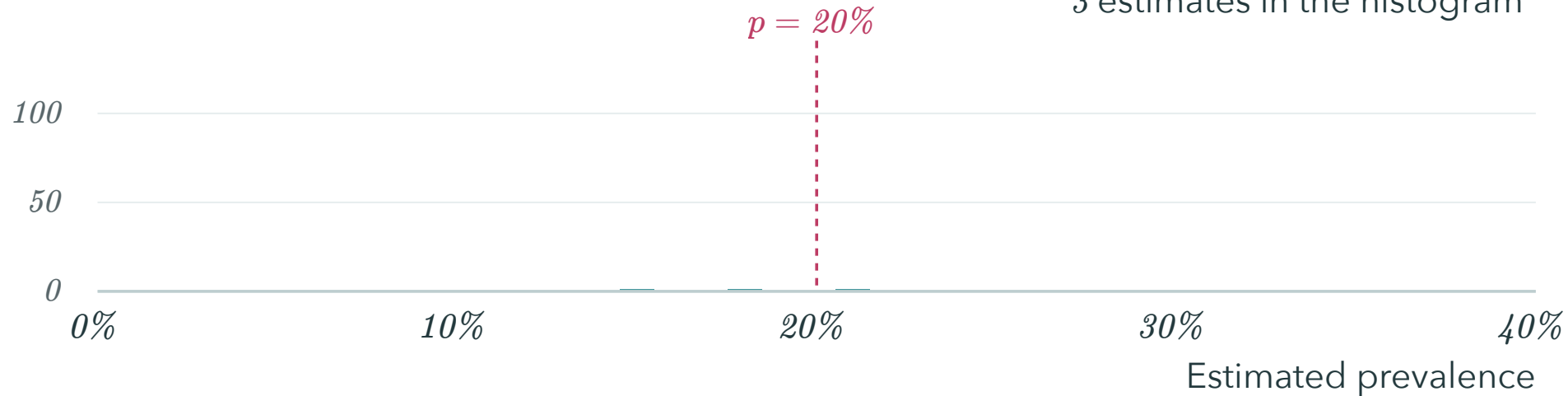
Latest sample: 3 ( $n = 100$ )



Estimate

$$\hat{p} = \frac{21}{100} = 21\%$$

One value enters the histogram  
3 estimates in the histogram



3 samples, 3 estimates

The population stays fixed. The estimate changes with the sample.

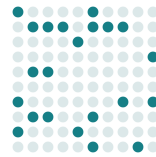
# The distribution of sample proportions

The same process continues: one sample, one estimate.

Population ( $p = 20\%$ )



Latest sample: 1,000 ( $n = 100$ )

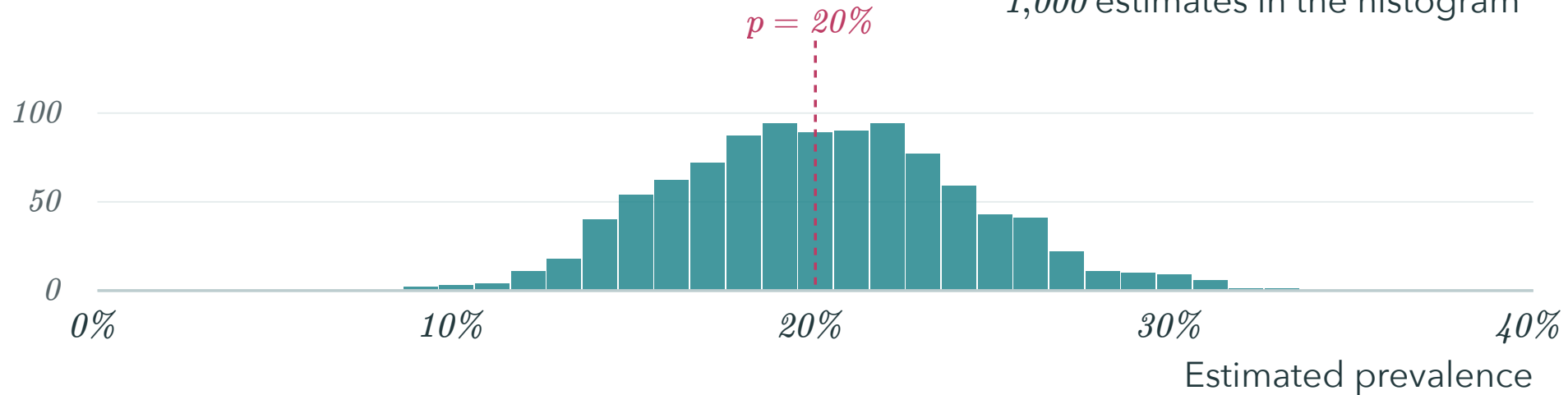


Estimate

$$\hat{p} = \frac{22}{100} = 22\%$$

One value enters the histogram

1,000 estimates in the histogram



1,000 samples, 1,000 estimates

The population stays fixed. The estimate changes with the sample.

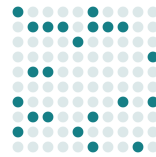
# The distribution of sample proportions

The same process continues: one sample, one estimate.

Population ( $p = 20\%$ )



Latest sample: 1,000 ( $n = 100$ )

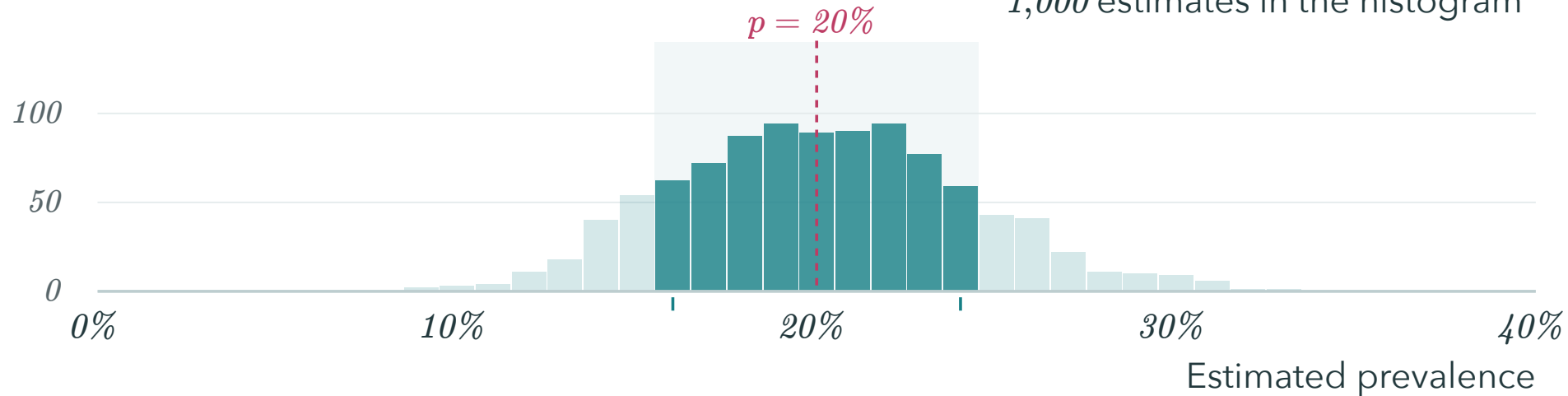


Estimate

$$\hat{p} = \frac{22}{100} = 22\%$$

One value enters the histogram

1,000 estimates in the histogram



724/1,000 estimates between 16% and 24% (72.4%)

A wider range around 20% contains more of the estimates.

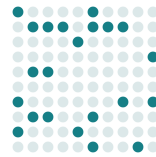
# The distribution of sample proportions

The same process continues: one sample, one estimate.

Population ( $p = 20\%$ )



Latest sample: 1,000 ( $n = 100$ )

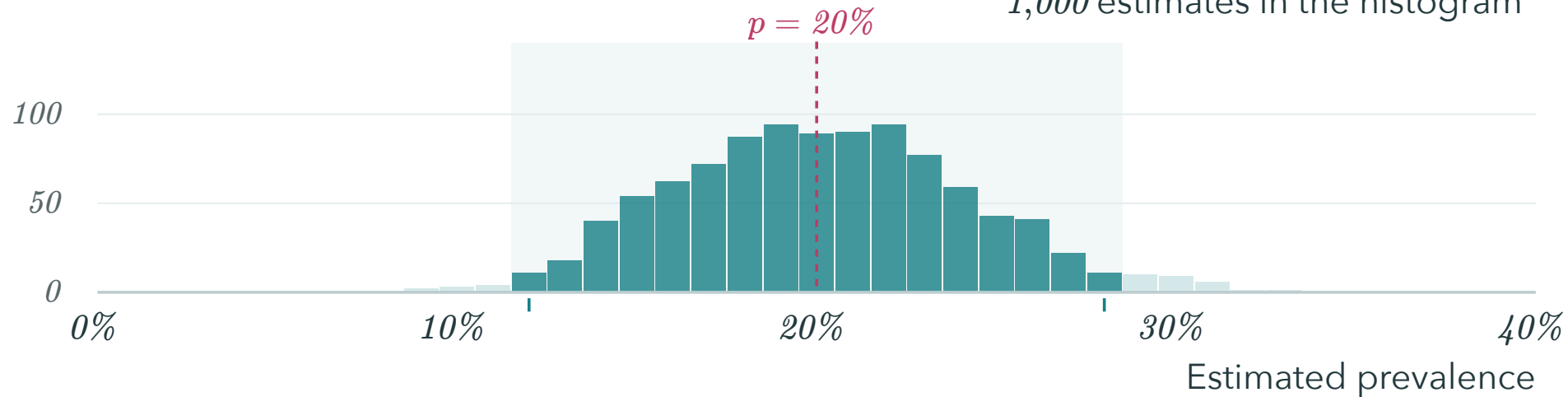


Estimate

$$\hat{p} = \frac{22}{100} = 22\%$$

One value enters the histogram

1,000 estimates in the histogram

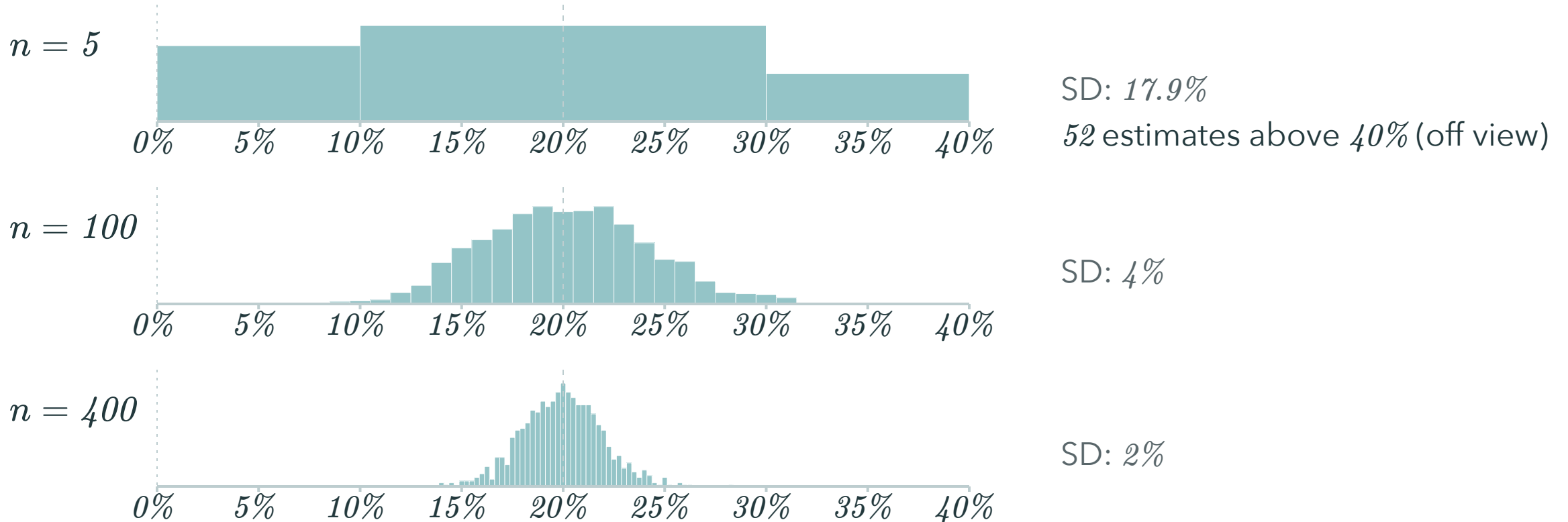


964/1,000 estimates between 12% and 28% (96.4%)

A wider range around 20% contains more of the estimates.

# Sample size and the normal approximation

Same prevalence: *20%*. Each row uses *1,000* estimates.

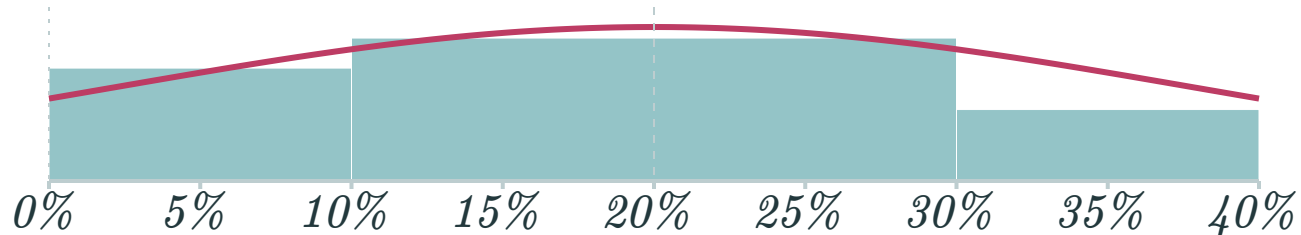


With 5 people, estimates are widely spaced. Larger samples give finer, narrower distributions.

# Sample size and the normal approximation

Same prevalence: 20%. Each row uses 1,000 estimates.

$n = 5$

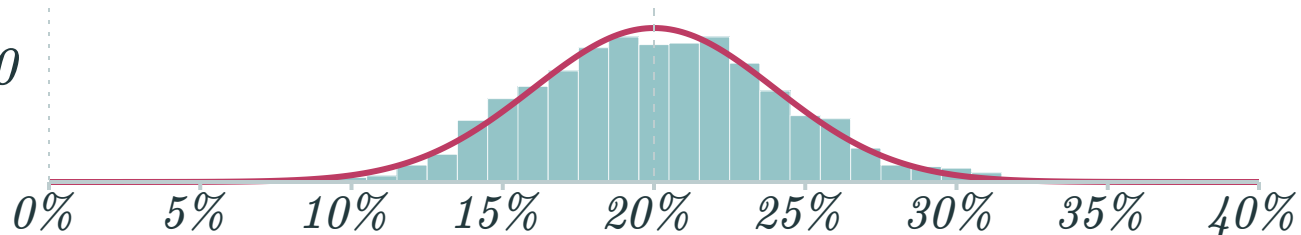


$$\mathcal{N}\left(0.20, \frac{0.16}{5}\right)$$

SD: 17.9%

52 estimates above 40% (off view)

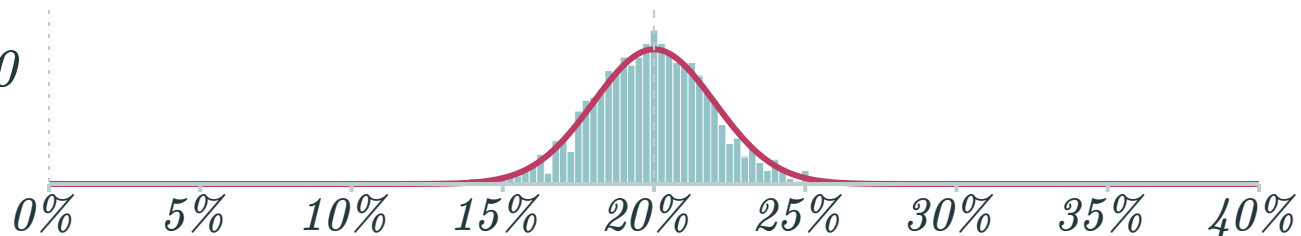
$n = 100$



$$\mathcal{N}\left(0.20, \frac{0.16}{100}\right)$$

SD: 4%

$n = 400$



$$\mathcal{N}\left(0.20, \frac{0.16}{400}\right)$$

SD: 2%

Rose: normal approximation, cropped at the edges. At  $n = 5$ , it is still a rough approximation.

# The central limit theorem and standard error

$\hat{p}$  averages  $n$  independent 0/1 observations with the same probability  $p$ .

For large  $n$  (people in each sample)

$$\hat{p} \stackrel{\text{approx.}}{\sim} \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

$$\text{Mean} = p$$

$$\text{Variance} = p(1-p)/n$$

$$\text{SE} = \sqrt{\frac{p(1-p)}{n}}$$

The standard error is the SD of the estimator across possible samples.

# The central limit theorem and standard error

$\hat{p}$  averages  $n$  independent 0/1 observations with the same probability  $p$ .

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$$\text{Variance} = p(1-p)/n$$

$$\text{SE} = \sqrt{\frac{p(1-p)}{n}}$$

| $p = 20\%$ | Exact variance of $\hat{p}$ | Exact SE |
|------------|-----------------------------|----------|
| $n = 25$   | 0.0064                      | 8%       |
| $n = 100$  | 0.0016                      | 4%       |
| $n = 400$  | 0.0004                      | 2%       |

Quadrupling the sample size halves the standard error.

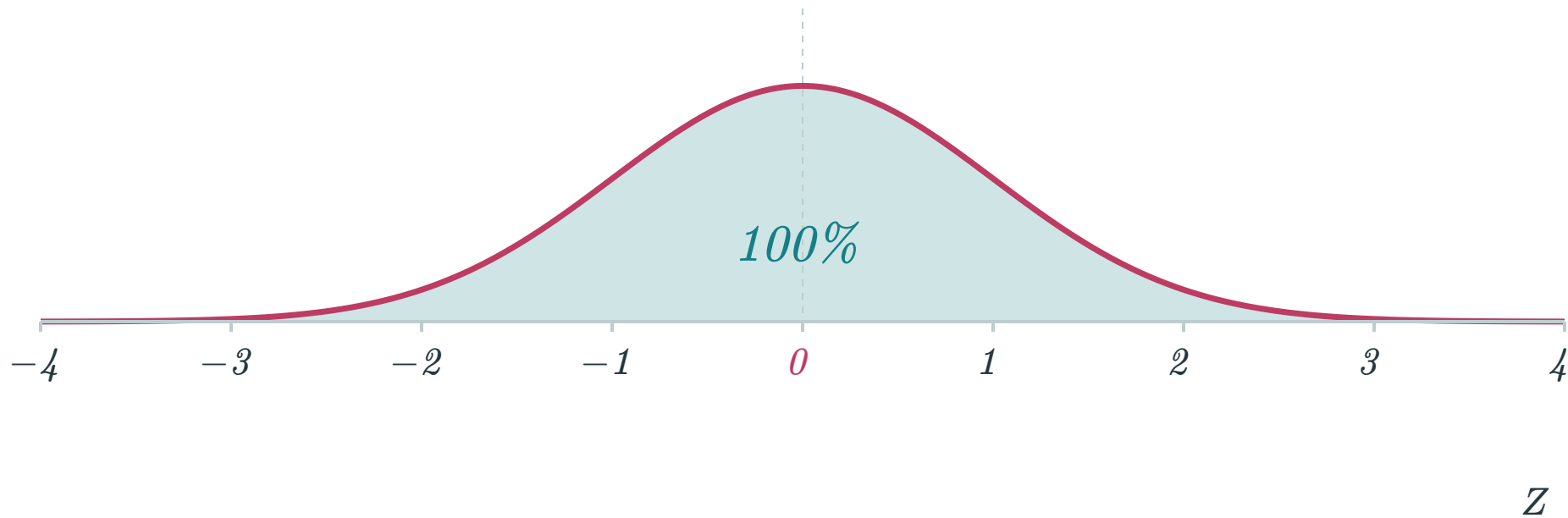
# The standard normal distribution

$$Z \sim \mathcal{N}(0, 1)$$

Mean  
*0*

Variance  
*1*

SD  
*1*



The total area represents *100%* probability.

$Z$  is a random variable. Areas under its curve describe probabilities.

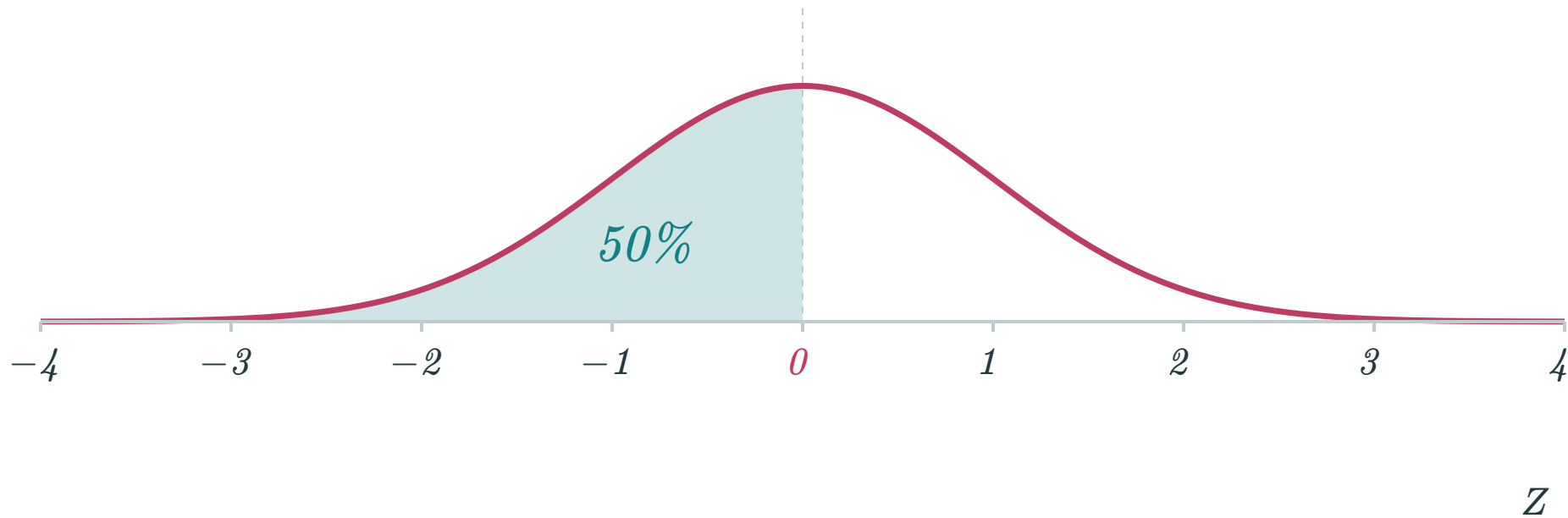
# The standard normal distribution

$$Z \sim \mathcal{N}(0, 1)$$

Mean  
*0*

Variance  
*1*

SD  
*1*



$$\Pr(Z < 0) = 0.5$$

The standard normal is symmetric. Its mean and median are both zero.

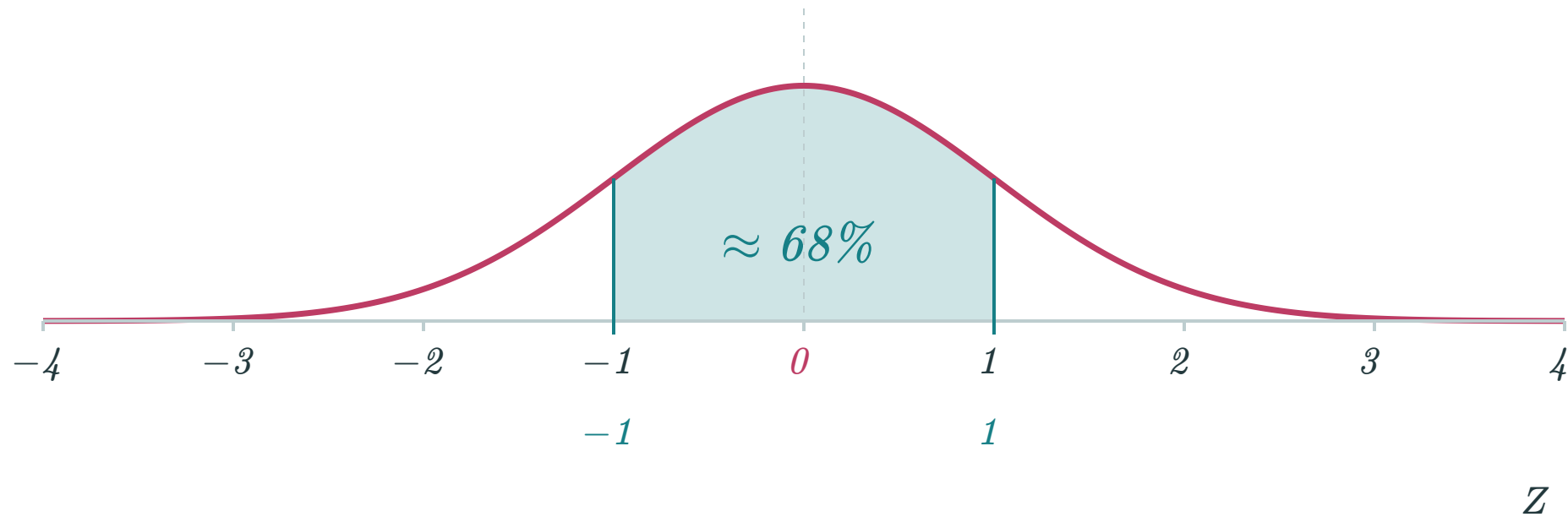
# The standard normal distribution

$$Z \sim \mathcal{N}(0, 1)$$

Mean  
*0*

Variance  
*1*

SD  
*1*



$$\Pr(-1 < Z < 1) \approx 0.68$$

About 68% probability of a value between  $-1$  and  $+1$ .

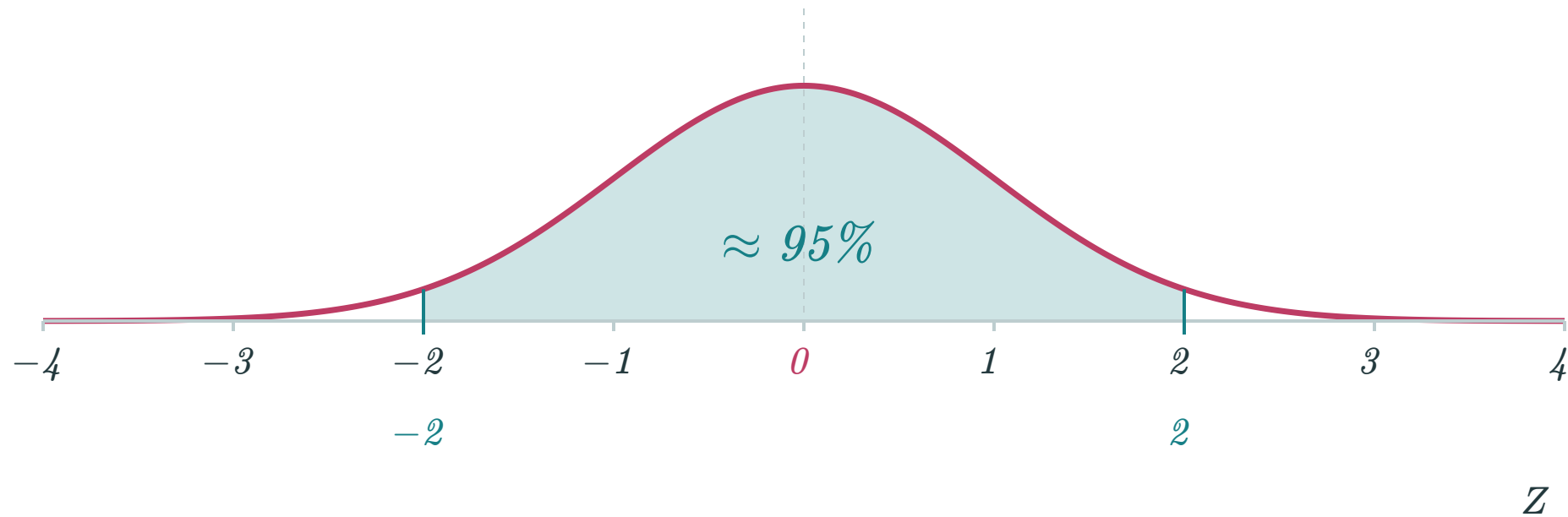
# The standard normal distribution

$$Z \sim \mathcal{N}(0, 1)$$

Mean  
*0*

Variance  
*1*

SD  
*1*



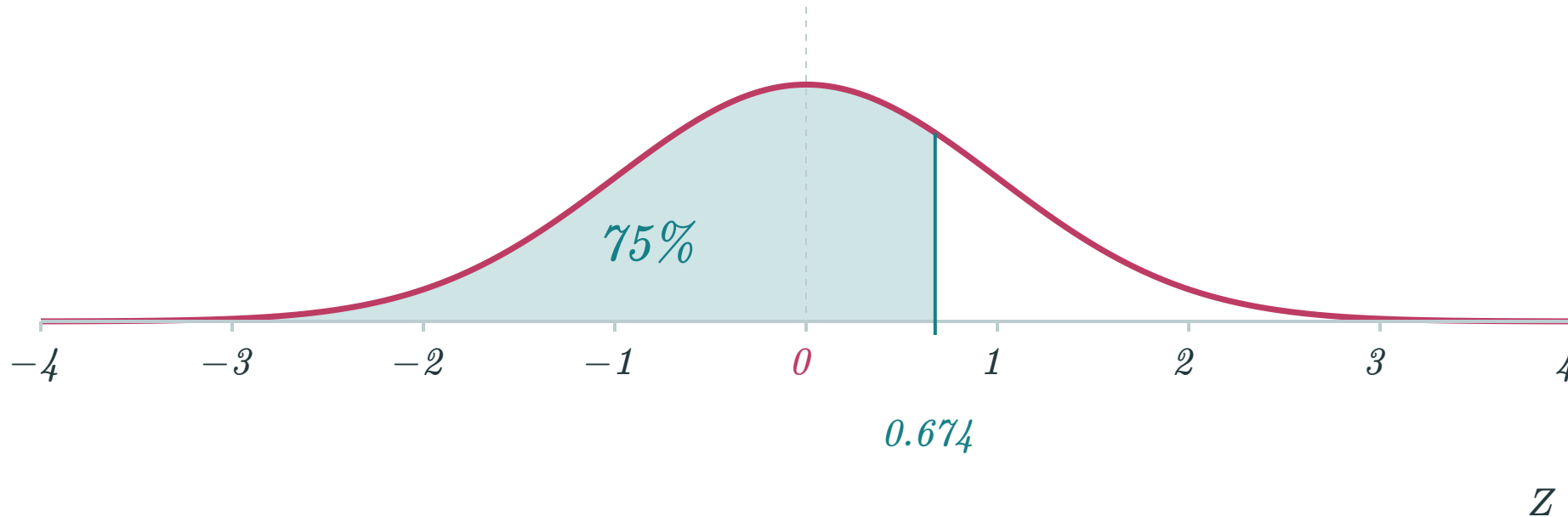
$$\Pr(-2 < Z < 2) \approx 0.9545$$

About 95% probability of a value between  $-2$  and  $+2$ .

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



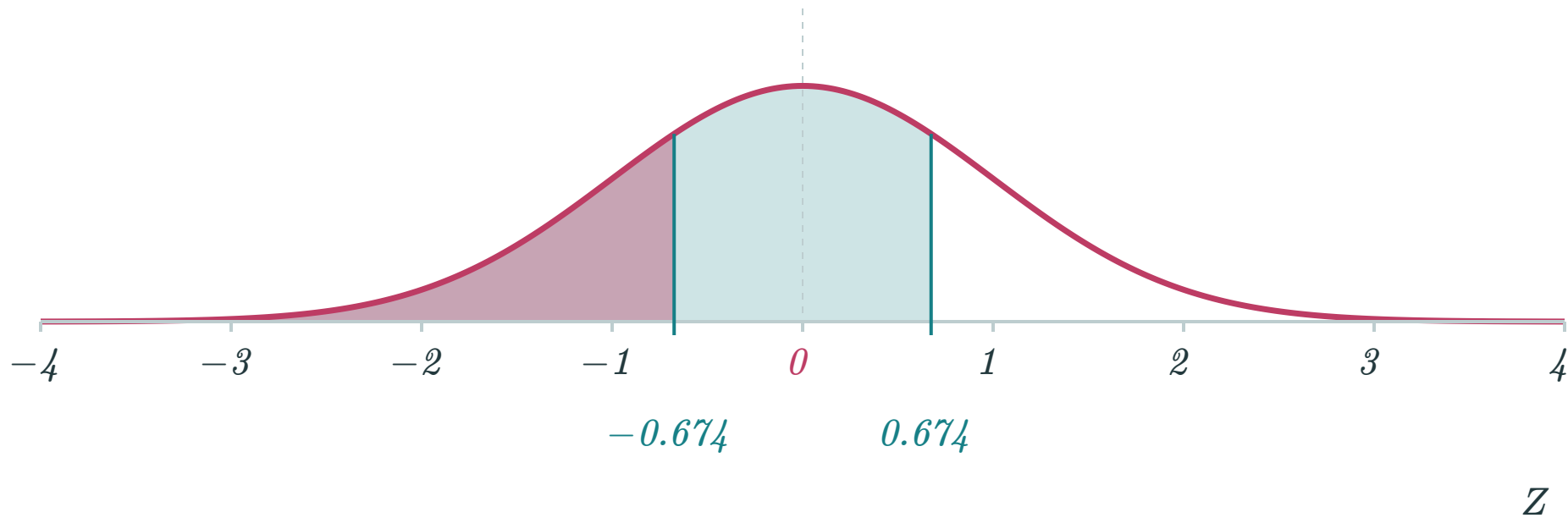
$$\Pr(Z \leq z_q) = q, \quad 0 < q < 1; \quad z_{0.75} \approx 0.674$$

$q$  is a probability. The quantile  $z_q$  is its cutoff on the horizontal axis.

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



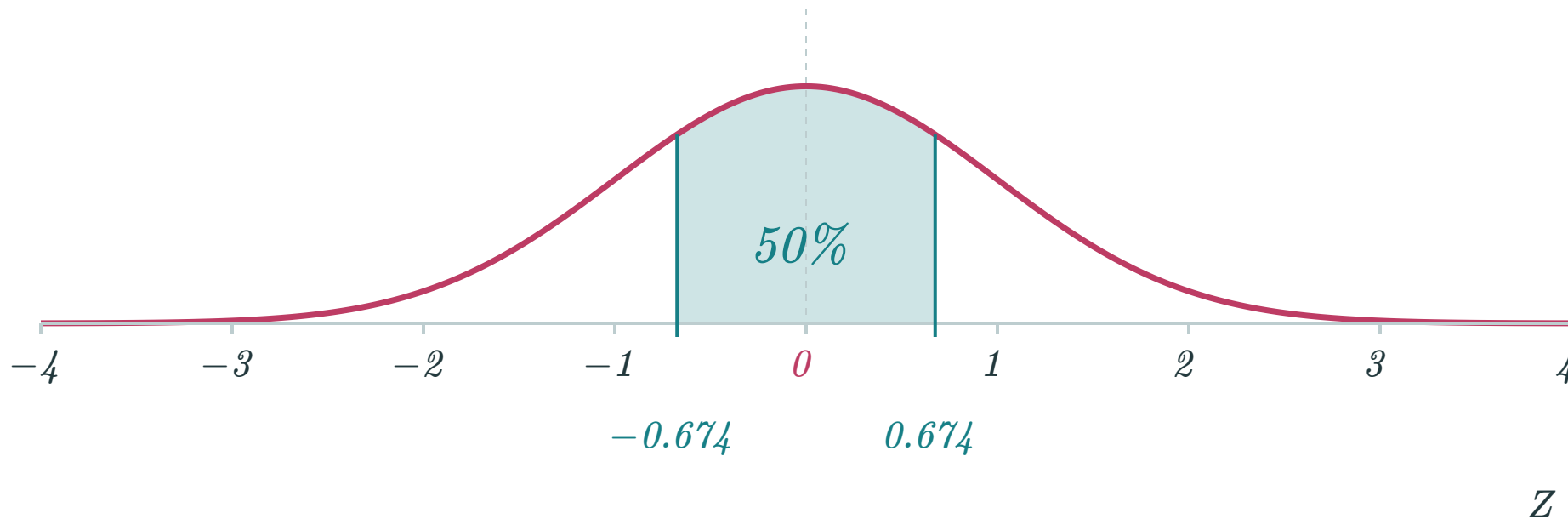
$$q_1 = 0.25, \quad q_2 = 0.75 \quad q_2 - q_1 = 0.50$$

Start with the 75% area. The rose area marks the 25% to subtract.

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



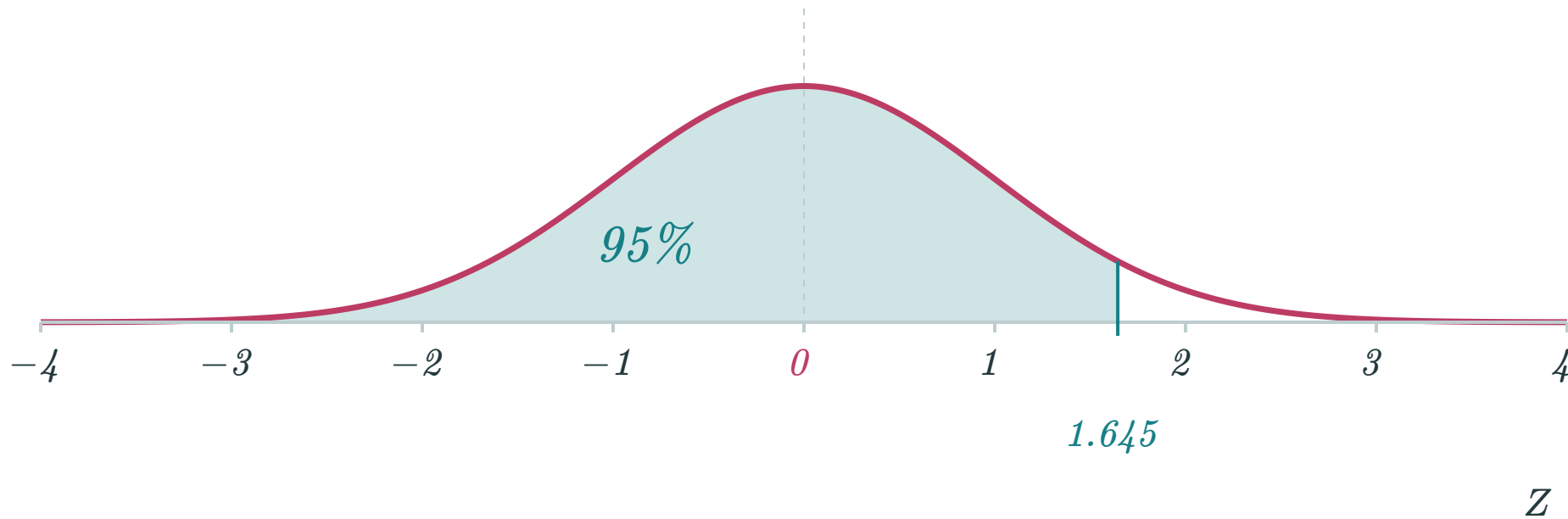
$$\Pr(z_{q_1} < Z \leq z_{q_2}) = q_2 - q_1$$

Subtracting the two cumulative areas leaves the area between the cutoffs.

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



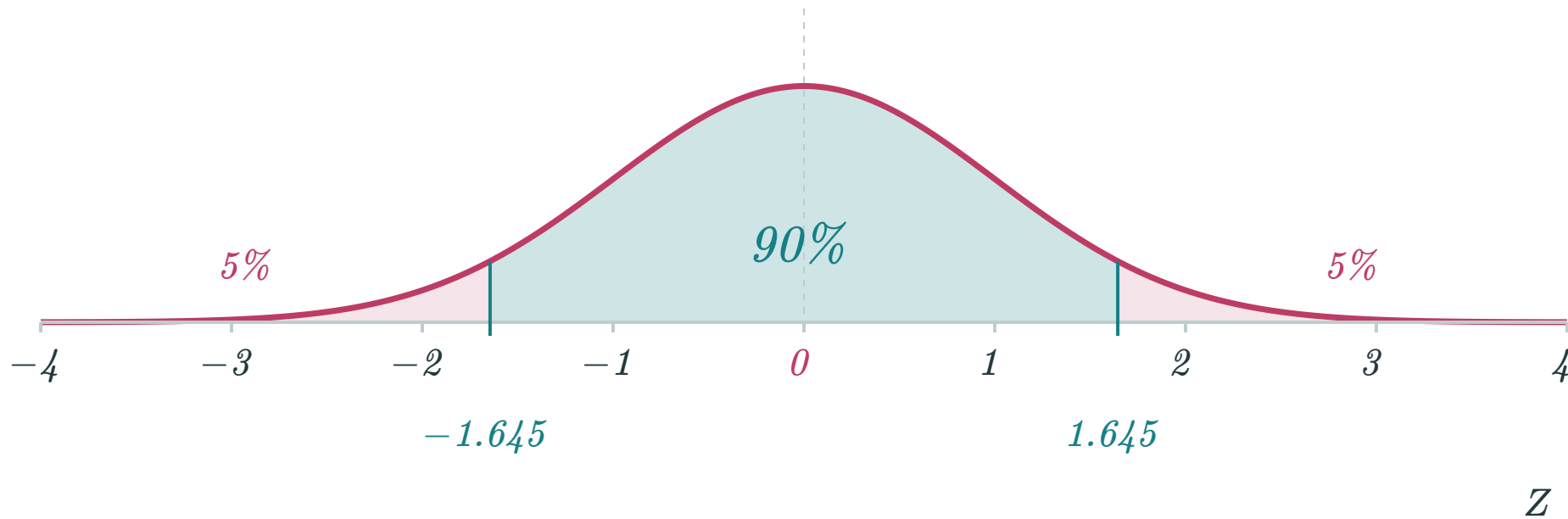
$$z_{0.95} \approx 1.645 \quad \Pr(Z \leq z_{0.95}) = 0.95$$

A 95% quantile leaves 5% on its right. It is about 1.645, not 1.96.

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



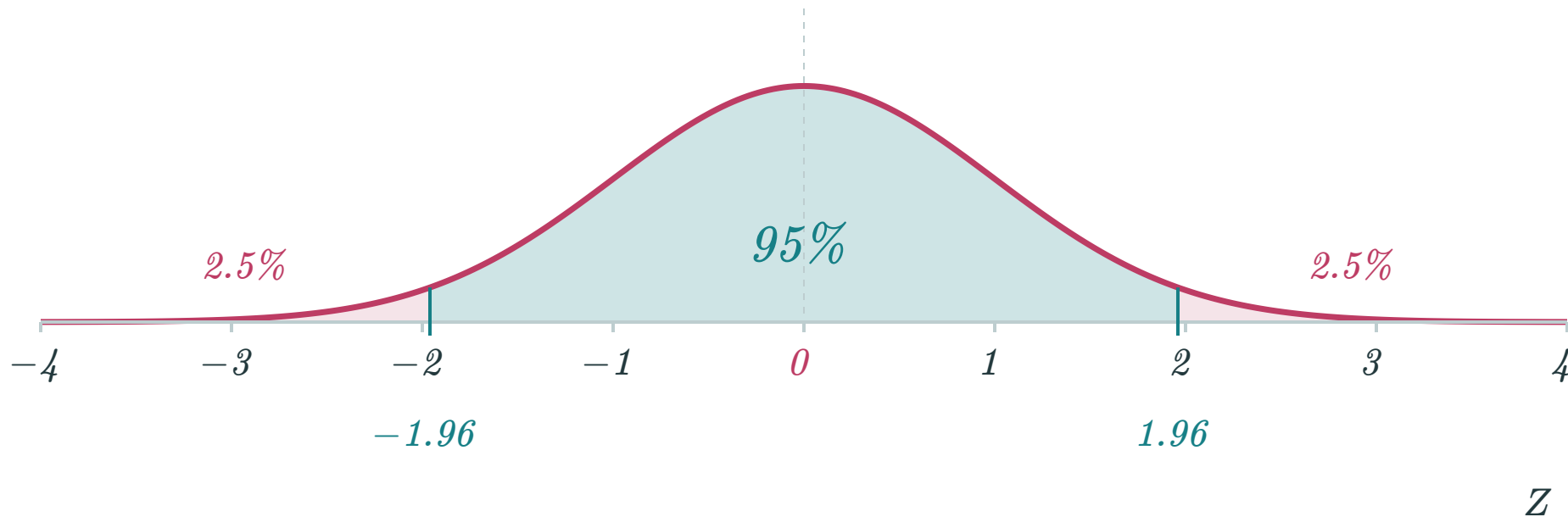
$$\Pr(z_{0.05} < Z < z_{0.95}) = 0.90$$

The 5% and 95% quantiles enclose the central 90%.

# Standard Normal quantiles

$$Z \sim \mathcal{N}(0, 1)$$

Quantiles from a table



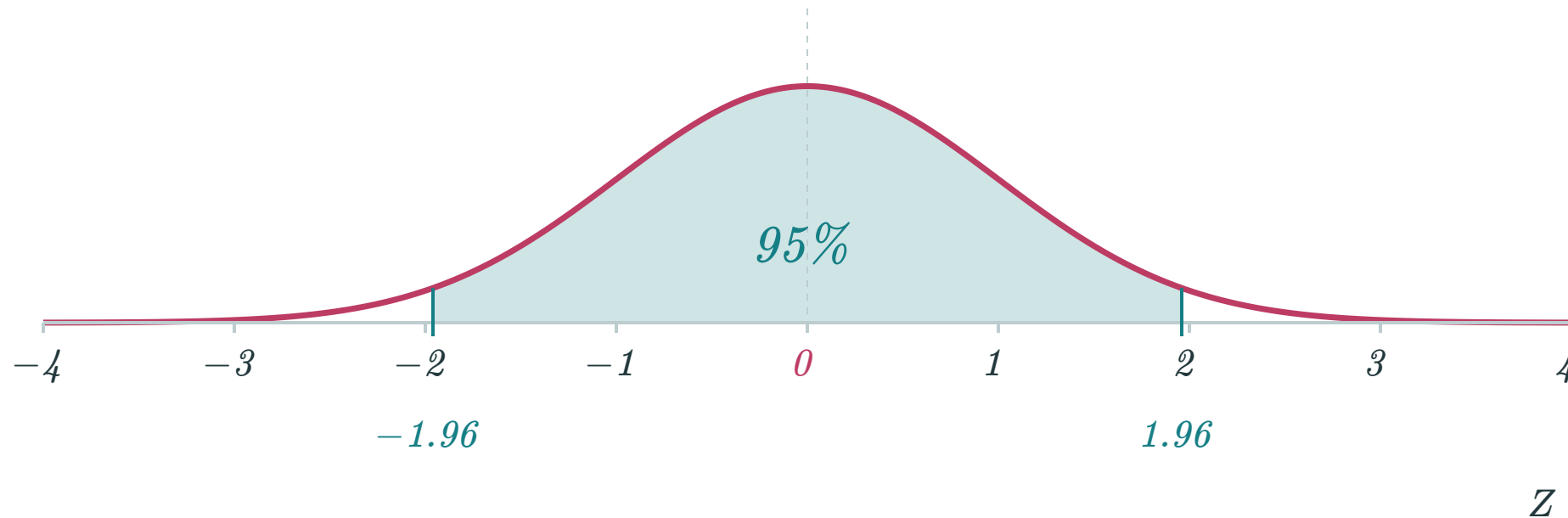
$$\Pr(z_{0.025} < Z < z_{0.975}) = 0.95$$

The central 95% uses the 2.5% and 97.5% quantiles, approximately  $-1.96$  and  $+1.96$ .

# The normal approximation on the prevalence scale

$$Z \sim \mathcal{N}(0, 1)$$

The central 95%



$$\Pr(-z_{0.975} < Z < z_{0.975}) = 0.95$$

Follow the centre and the two cutoffs.

# The normal approximation on the prevalence scale

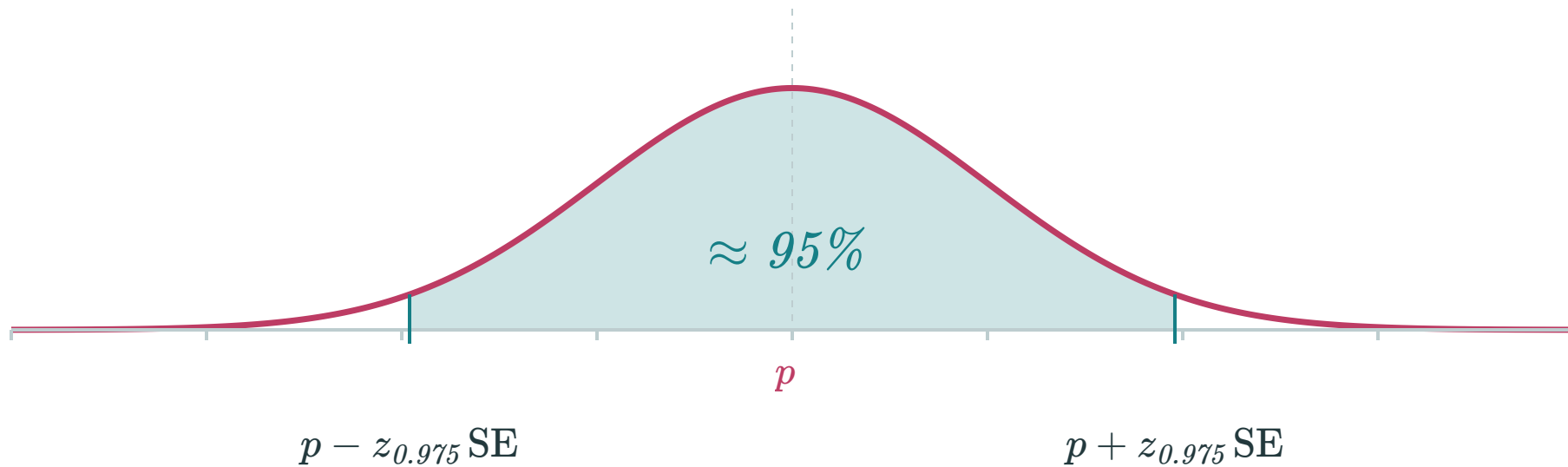
Standard reference

$$Z \sim \mathcal{N}(0, 1)$$



Sampling approximation

$$\hat{p} \overset{\text{approx.}}{\sim} \mathcal{N}(p, \text{SE}^2)$$



Sample proportion

$$\Pr(p - z_{0.975}\text{SE} \leq \hat{p} \leq p + z_{0.975}\text{SE}) \approx 0.95$$

Multiply by the standard error, then add the population prevalence.

# The normal approximation on the prevalence scale

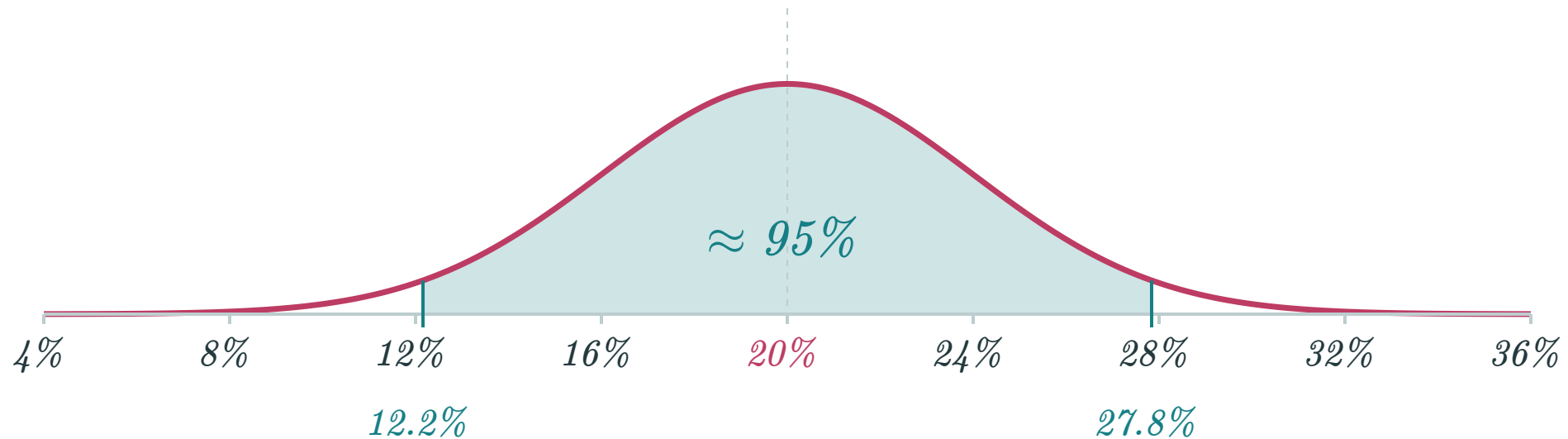
Standard reference

$$Z \sim \mathcal{N}(0, 1)$$



Sampling approximation

$$\hat{p} \overset{\text{approx.}}{\sim} \mathcal{N}(p, \text{SE}^2)$$



Estimated prevalence

$$\Pr(12.16\% \leq \hat{p} \leq 27.84\%) \approx 0.95$$

A range of possible estimates around the known population value.

# The normal approximation on the prevalence scale

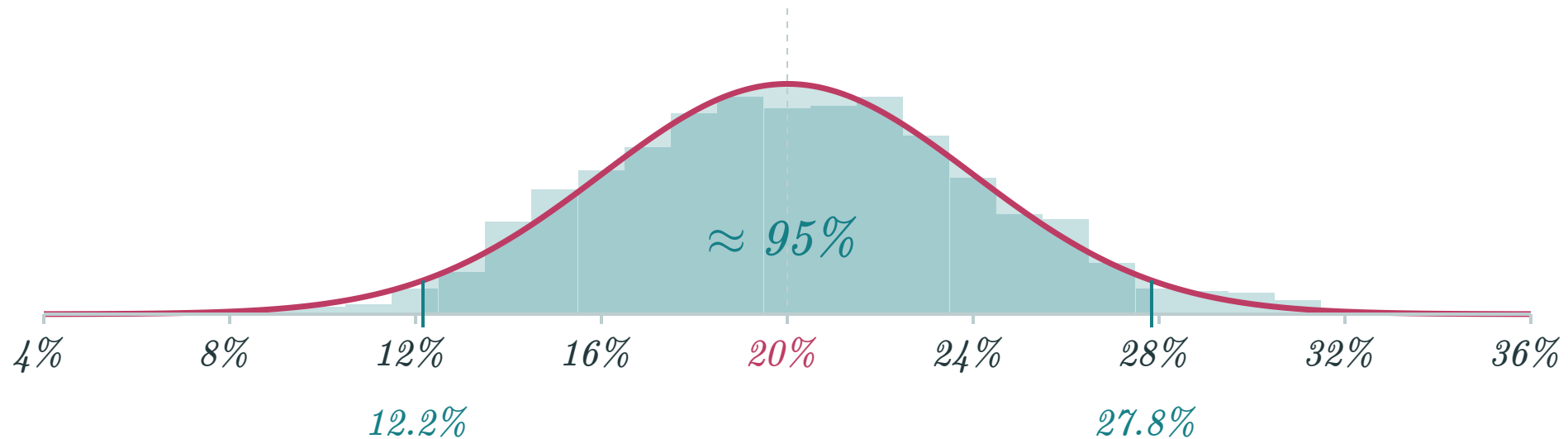
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Sampling approximation

$$\hat{p} \stackrel{\text{approx.}}{\sim} \mathcal{N}(p, \text{SE}^2)$$



Estimated prevalence

$$\Pr(12.16\% \leq \hat{p} \leq 27.84\%) \approx 0.95$$

942/1,000 simulated estimates fall in this range. The histogram is only approximately normal.

# In practice, $p$ is unknown

We have one sample: *18* cases among *100* people.

Estimate the prevalence

$$\hat{p} = 18/100 = 0.18$$

Estimate its standard error

$$\widehat{\text{SE}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

---

The unknown population value is replaced by an estimate from the sample.

# In practice, $p$ is unknown

We have one sample:  $18$  cases among  $100$  people.

Estimate the prevalence

$$\hat{p} = 18/100 = 0.18$$

Estimate its standard error

$$\widehat{\text{SE}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

---

$$\widehat{\text{SE}} = \sqrt{\frac{0.18 \times 0.82}{100}} \approx 0.0384 = 3.84\%$$

In our simulation, this estimates  $4\%$ , just as  $18\%$  estimates  $20\%$ .

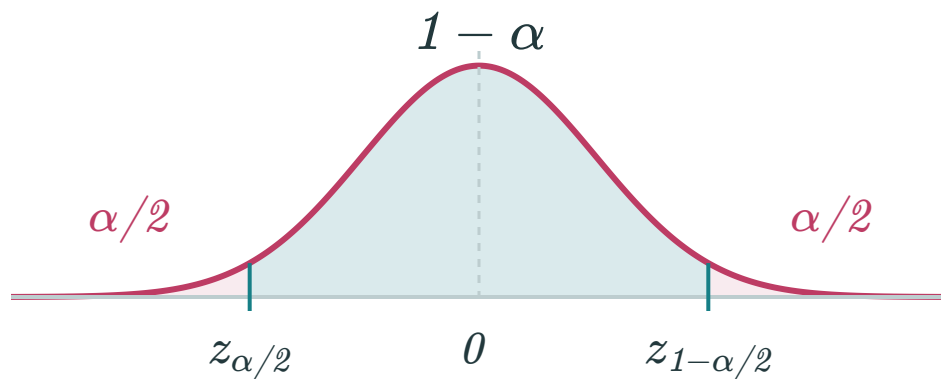
# Confidence interval and its margin of error

Let  $0 < \alpha < \frac{1}{2}$ . Confidence level:  $1 - \alpha$ .

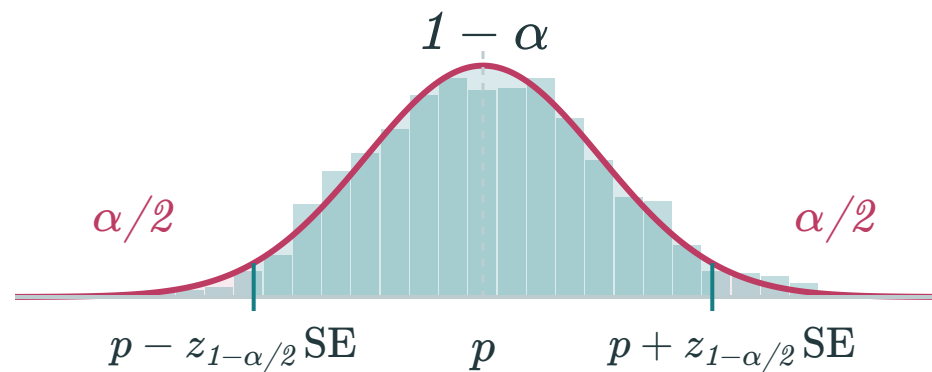
$$\text{CI}_{1-\alpha} : \hat{p} \pm E \quad E = z_{1-\alpha/2} \widehat{\text{SE}}$$

The margin of error is the half-width of the interval.

$$Z \sim \mathcal{N}(0, 1)$$



$$\hat{p} \stackrel{\text{approx.}}{\sim} \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$



The sampling distribution is centred on  $p$ .

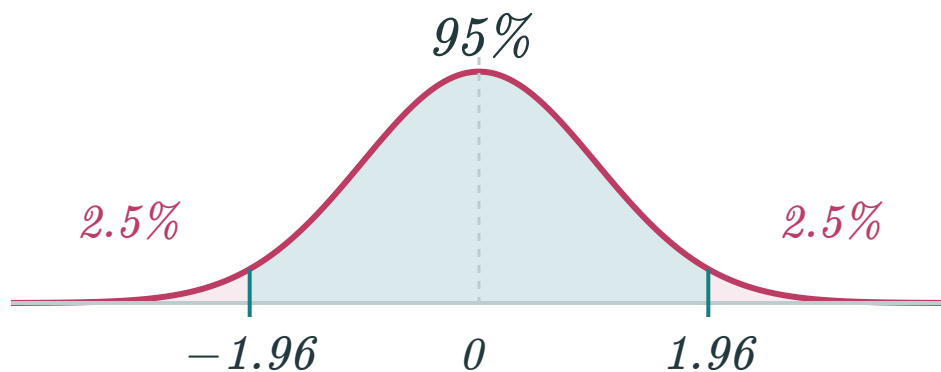
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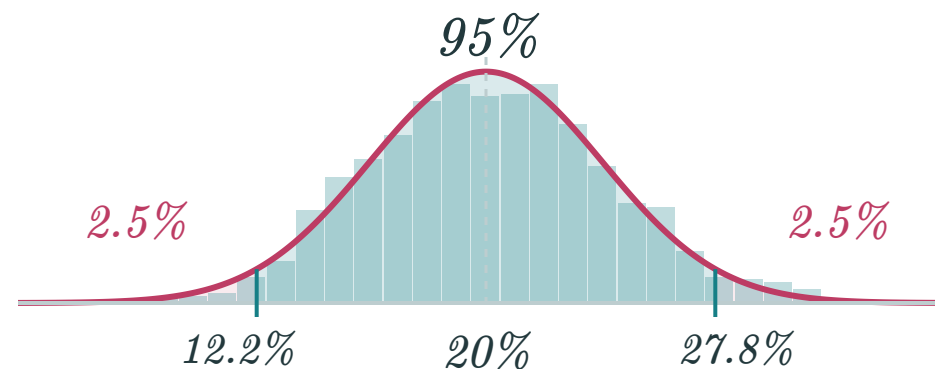
$$\text{CI}_{1-\alpha} : \hat{p} \pm E \quad E = z_{1-\alpha/2} \widehat{\text{SE}}$$

95% confidence:  $\alpha = 0.05$ ,  $z_{0.975} \approx 1.96$

$$Z \sim \mathcal{N}(0, 1)$$



$$\hat{p} \stackrel{\text{approx.}}{\sim} \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$



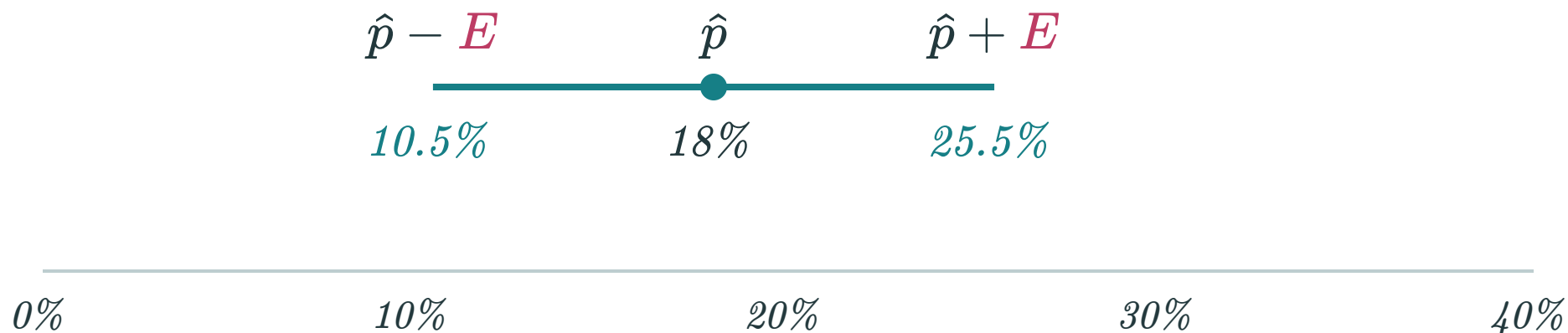
The sampling distribution is centred on  $p$ .

# Confidence interval and its margin of error

One sample: 18 cases among 100 people.

95% confidence:  $\alpha = 0.05$ ,  $z_{0.975} \approx 1.96$ .

$$E = 1.96 \times 3.84\% \approx 7.53\%$$

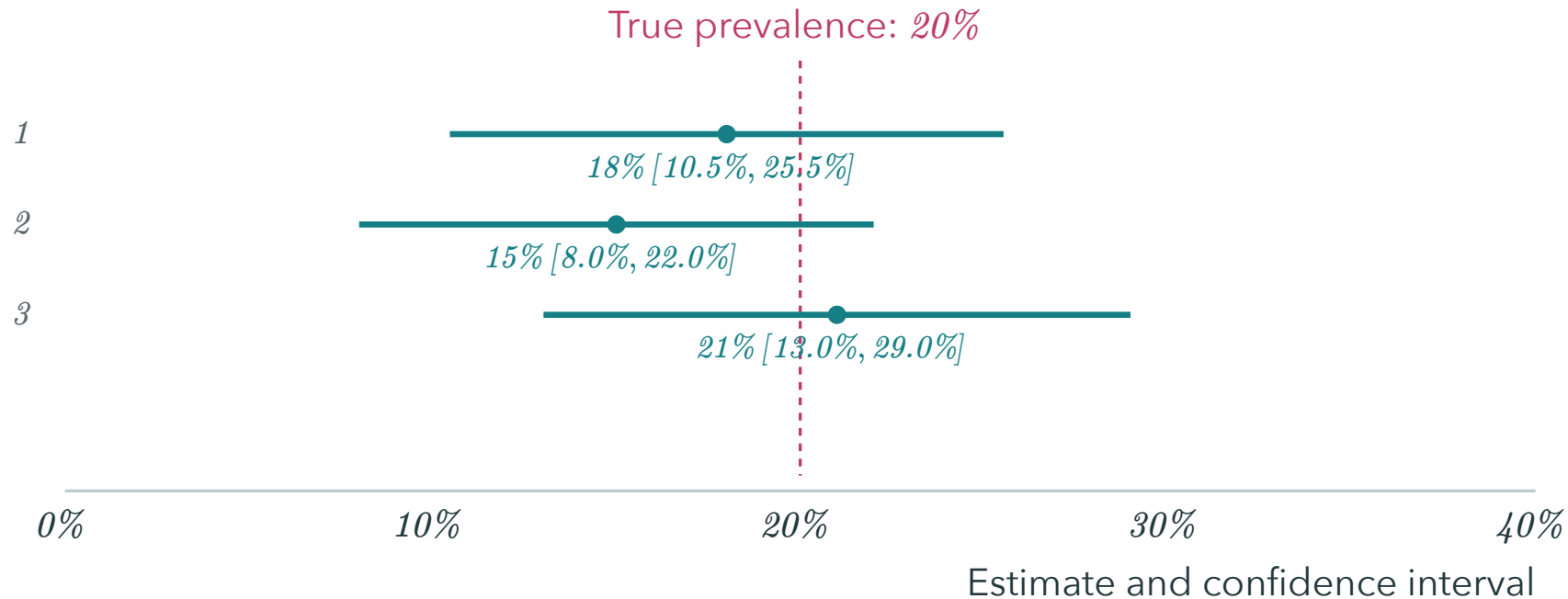


The margin of error is the half-width of the interval.

# Confidence intervals in repeated samples

Sample size 100

Confidence level 95%



*3/3 intervals contain 20% (100.0%)*

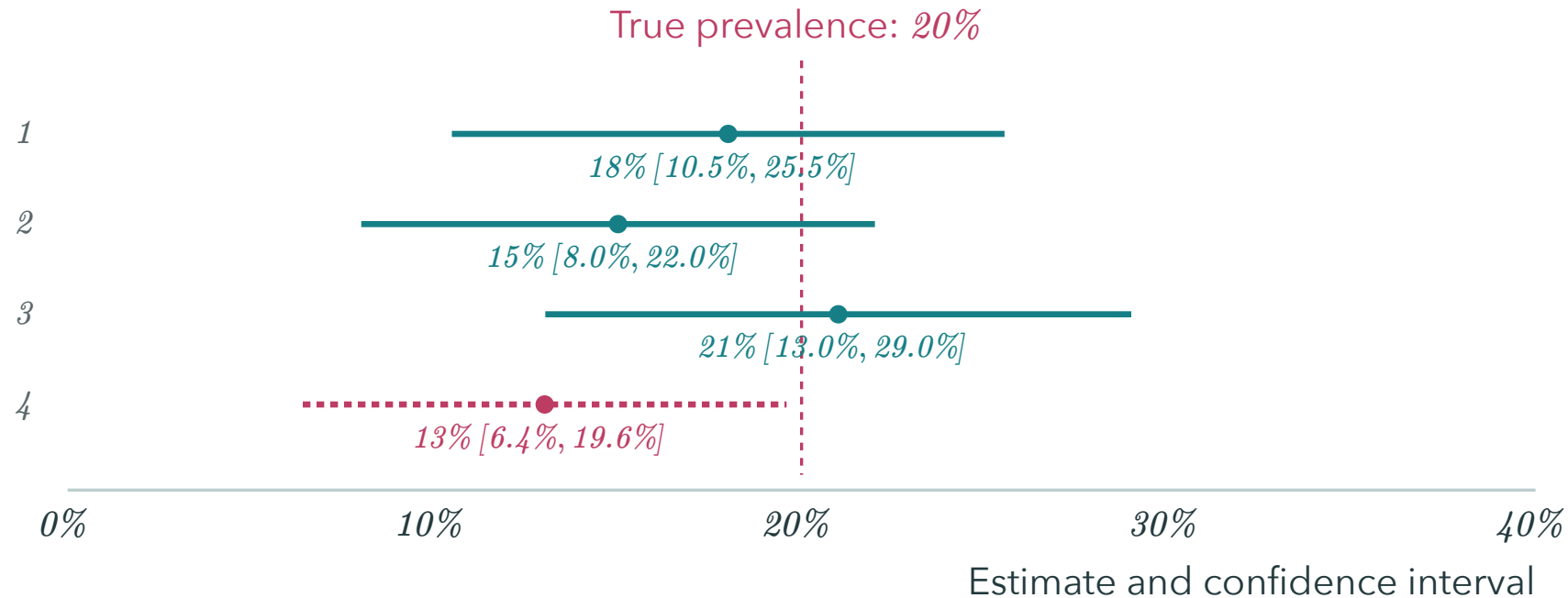
Same seeded samples as before

Each new sample gives a new estimate and interval.

# Confidence intervals in repeated samples

Sample size 100

Confidence level 95%



$3/4$  intervals contain 20% (75.0%)

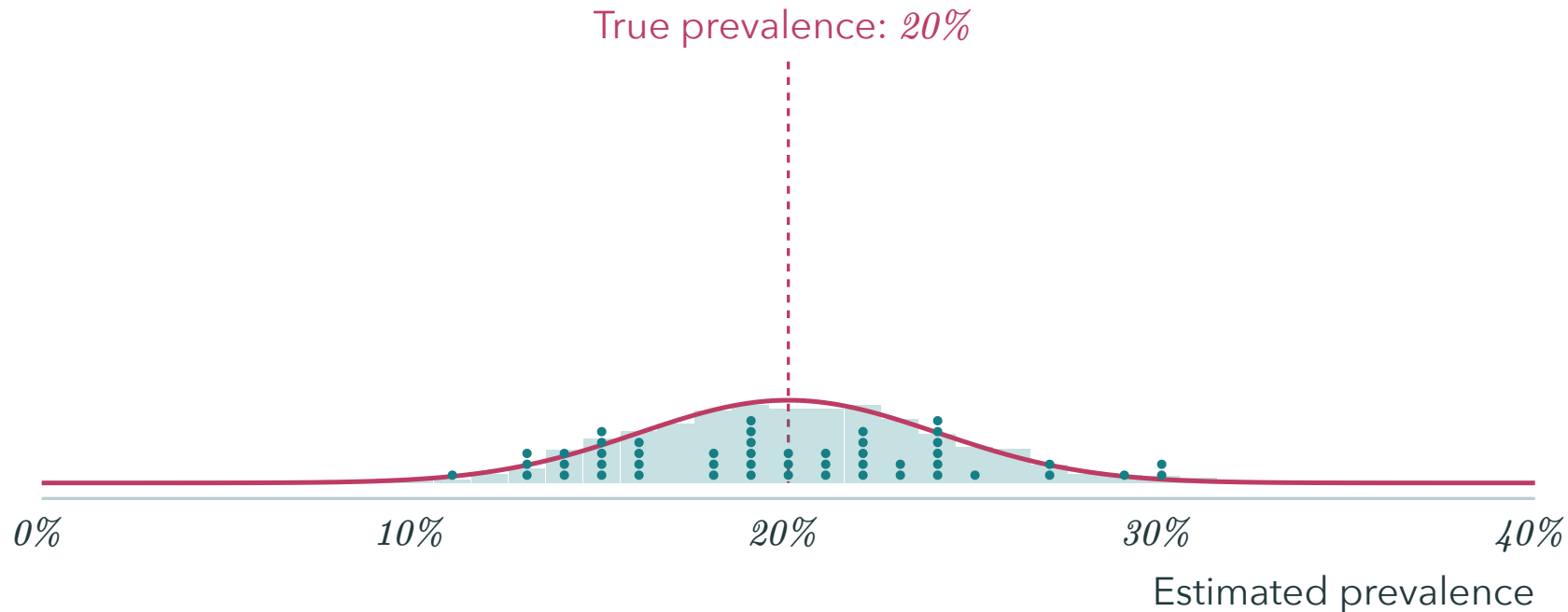
Same seeded samples as before

Four samples do not determine the long-run coverage rate.

# Confidence intervals in repeated samples

Sample size 100

Confidence level 95%



The same estimates form the histogram; 50 points are highlighted.

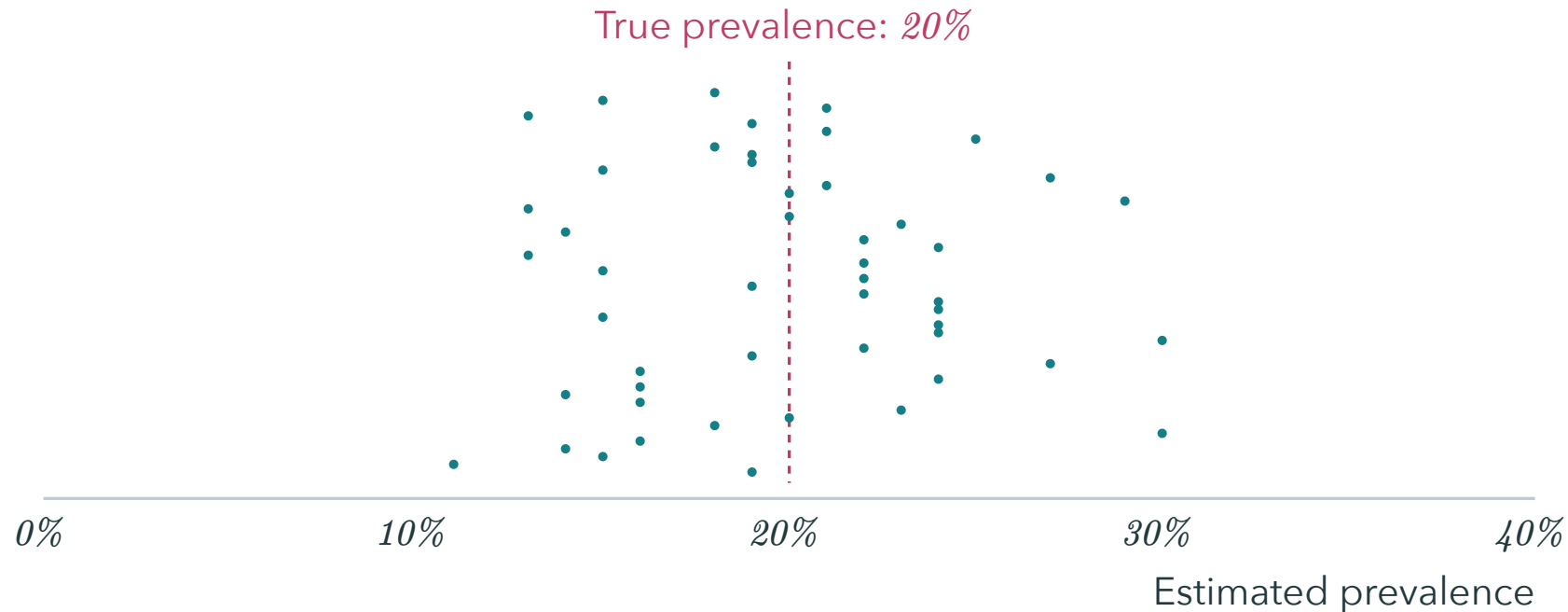
Rows 1-50 shown; 1,000 counted

Every point represents one sample. The true prevalence stays at 20%.

# Confidence intervals in repeated samples

Sample size 100

Confidence level 95%



Spread the same points into rows. Their estimates stay unchanged.

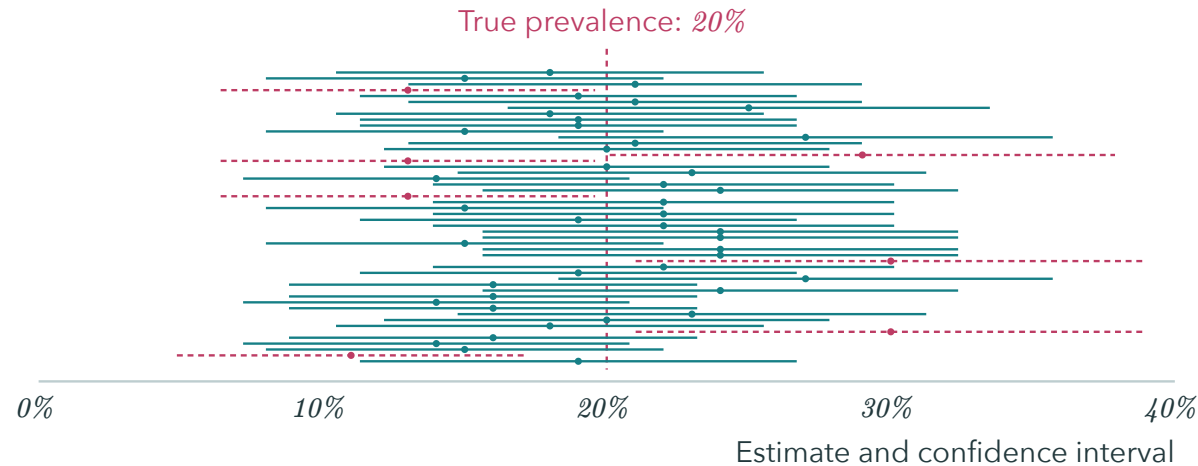
Rows 1-50 shown; 1,000 counted

Every point represents one sample. The true prevalence stays at 20%.

# Confidence intervals in repeated samples

Sample size 100

Confidence level 95%



*935/1,000 intervals contain 20% (93.5%)*

Rows 1-50 shown; 1,000 counted

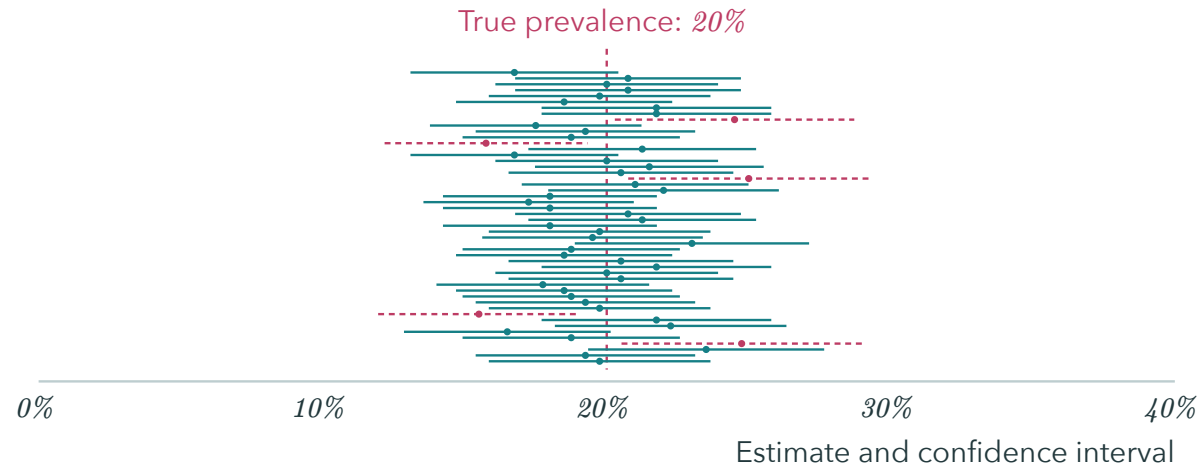
*≈ 95%* of intervals contain the true value  
across repeated samples.  
For a well-calibrated 95% confidence procedure.

Approximate coverage. The observed fraction can differ from 95%. Arrows mark off-scale ends.

# Confidence intervals in repeated samples

Sample size 400

Confidence level 95%



*938/1,000* intervals contain 20% (*93.8%*)

Rows 1-50 shown; 1,000 counted

$\approx 95\%$

of intervals contain the true value  
across repeated samples.

For a well-calibrated 95% confidence procedure.

Approximate coverage. The observed fraction can differ from 95%. Arrows mark off-scale ends.

# What would a *100%* confidence interval be?

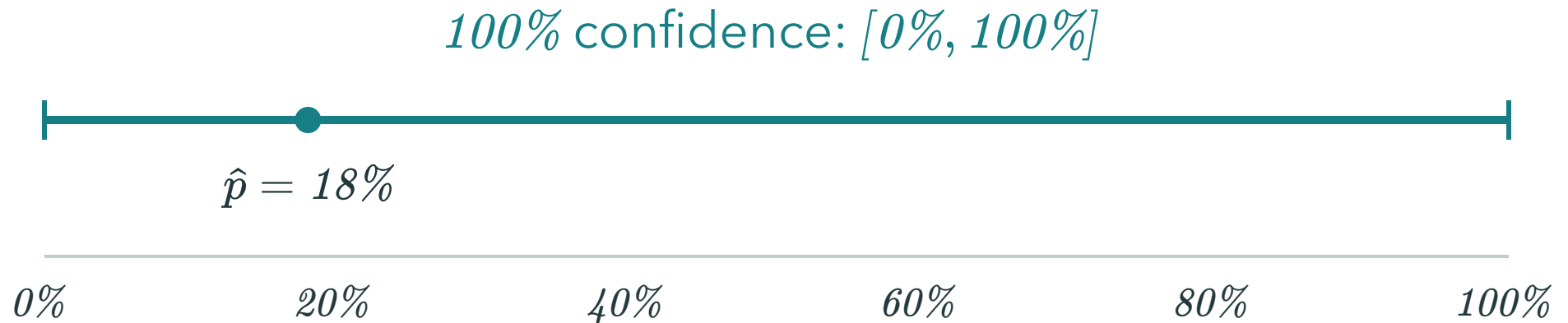
What if we want to be sure to include the true proportion?



We have one observed estimate. How wide must the interval be?

# What would a *100%* confidence interval be?

What if we want to be sure to include the true proportion?

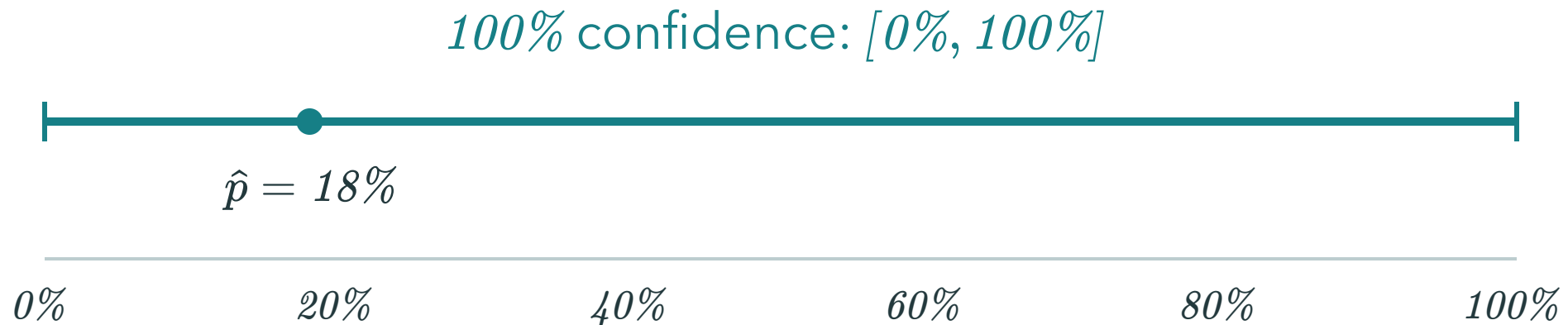


Every proportion lies between *0%* and *100%*.

This interval always contains the true proportion.

# What would a *100%* confidence interval be?

What if we want to be sure to include the true proportion?



Every proportion lies between *0%* and *100%*.

We need no data to say this. It does not tell us where the true value lies.

With the same data, confidence and precision involve a trade-off.

# Confidence intervals for a population mean

Systolic blood pressure in *100* adults

We want to estimate the population mean  $\mu$ .

Sample mean

$$\bar{x} = \frac{1}{100} \sum_{i=1}^{100} x_i = 125 \text{ mmHg}$$

Sample standard  
deviation

$$\hat{\sigma} = \sqrt{\frac{1}{100 - 1} \sum_{i=1}^{100} (x_i - \bar{x})^2} = 15 \text{ mmHg}$$

---

SD describes individual values. SE describes the uncertainty of the estimated mean.

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Estimated standard error of the mean

$$\widehat{\text{SE}} = \frac{\hat{\sigma}}{\sqrt{n}} = \frac{15}{\sqrt{100}} = 1.5 \text{ mmHg}$$

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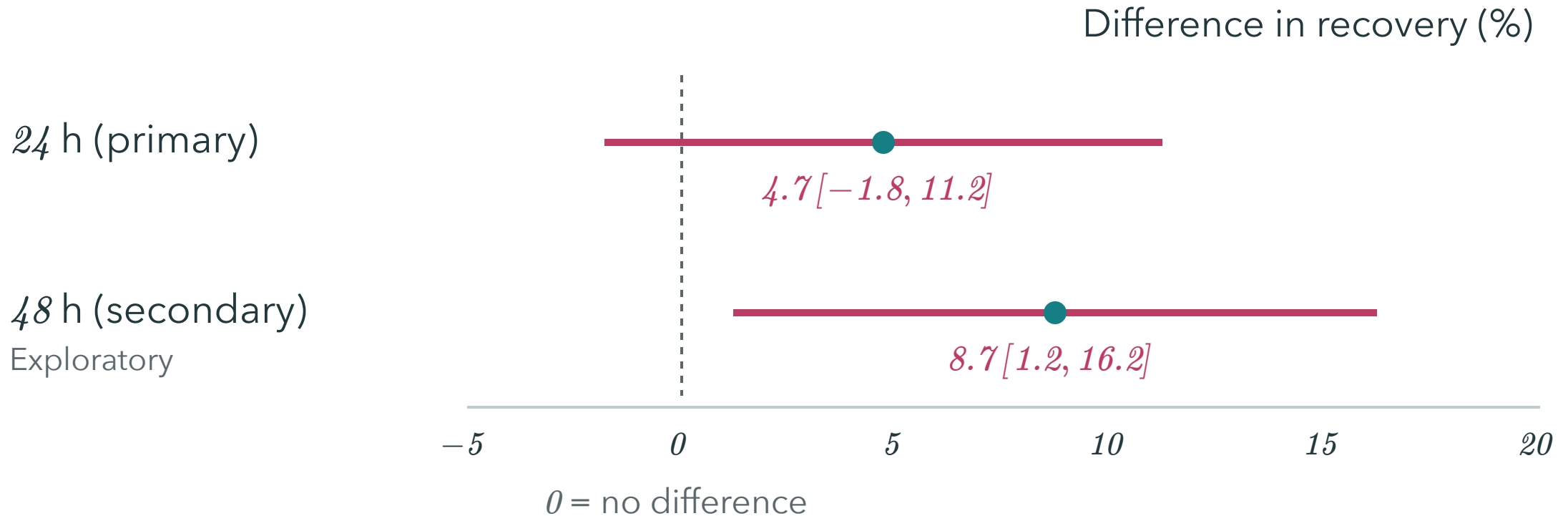
Approximate *95%* confidence interval

$$125 \pm 1.96 \times 1.5 \quad : \quad [122.1, 127.9] \text{ mmHg}$$

$E = 2.94$  mmHg. The same interval construction applies to a mean.

# Reading the JAMA confidence intervals

Dexamethasone versus placebo: adjusted differences in recovery.



Source: Hayward et al., *JAMA* 2017;317:1535-1543, Figure 2 (redrawn).

The point estimates the difference. The interval describes its precision.  
For a difference, 0 means no effect.

# Estimation and a new question

A *95%* confidence procedure covers the true value in about *95%* of repetitions, when well calibrated.

For a particular interval, the true value may or may not be inside.

The guarantee concerns the procedure across repeated samples.

# Estimation and a new question

A *95%* confidence procedure covers the true value in about *95%* of repetitions, when well calibrated.

For a particular interval, the true value may or may not be inside.

What if we want to test a guess on the true prevalence?

Hypothesis tests address this question using *p*-values.

Tests also rely on the sampling model to control error rates.

# References

1. Hayward G, Hay AD, Moore MV, et al. Effect of oral dexamethasone without immediate antibiotics vs placebo on acute sore throat in adults: a randomized clinical trial. *JAMA* 2017;317(15):1535. doi:[10.1001/jama.2017.3417](https://doi.org/10.1001/jama.2017.3417)

Thank you very much for your attention